

Lab 2 – 18.03.2026

During the class, we will follow a course that has been proposed by Google:

<https://developer.android.com/courses/android-basics-compose/course>

Tasks for the second lab:

1. Unit 2: Building App UI

<https://developer.android.com/courses/android-basics-compose/unit-2>

2. Pathway 1: Kotlin Fundamentals (*recommended, not mandatory*)

<https://developer.android.com/courses/pathways/android-basics-compose-unit-2-pathway-1>

Learn more fundamentals of Kotlin including conditionals, nullability, object-oriented programming, and lambda expressions.

a. Write Conditionals in Kotlin

<https://developer.android.com/codelabs/basic-android-kotlin-compose-conditionals>

b. Use Nullability in Kotlin

<https://developer.android.com/codelabs/basic-android-kotlin-compose-nullability>

c. Use Classes and Objects in Kotlin

<https://developer.android.com/codelabs/basic-android-kotlin-compose-classes-and-objects>

d. Use Function Types and Lambda Expressions in Kotlin

<https://developer.android.com/codelabs/basic-android-kotlin-compose-function-types-and-lambda>

e. Practice: Kotlin Fundamentals (**mandatory**)

<https://developer.android.com/codelabs/basic-android-kotlin-compose-kotlin-fundamentals-practice-problems>

3. Pathway 2: Add a Button to an App

<https://developer.android.com/courses/pathways/android-basics-compose-unit-2-pathway-2>

Learn how to respond to button clicks in an Android app by building an interactive Dice Roller application.

a. Create an interactive Dice Roller app

<https://developer.android.com/codelabs/basic-android-kotlin-compose-build-a-dice-roller-app>

- b. Use the debugger in Android Studio
<https://developer.android.com/codelabs/basic-android-kotlin-compose-intro-debugger>
- c. Practice: Click behavior
<https://developer.android.com/codelabs/basic-android-kotlin-compose-button-click-practice-problem>

4. Pathway 3: Interact with UI and State

<https://developer.android.com/courses/pathways/android-basics-compose-unit-2-pathway-3>

Learn how to manage state in Jetpack Compose by building a Tip Calculator app, then apply everything in a larger capstone project.

- a. Intro to state in Compose
<https://developer.android.com/codelabs/basic-android-kotlin-compose-using-state>
- b. Calculate a custom tip
<https://developer.android.com/codelabs/basic-android-kotlin-compose-calculate-tip>
- c. Write automated tests
<https://developer.android.com/codelabs/basic-android-kotlin-compose-write-automated-tests>
- d. Project: Create an Art Space app (**mandatory**)
<https://developer.android.com/codelabs/basic-android-kotlin-compose-art-space>

As a result of the work please take the Quiz at the end of each Pathway.

Submission requirements:

- Create a **private** repository named MAD on any Git hosting platform (e.g. GitHub, GitLab, Bitbucket). This repository will be used throughout the entire course – each unit will have its own folder inside it.
- Inside the repository maintain the following folder structure:

```
MAD/  
  UNIT2/  
    KotlinFundamentals/    (Practice: Kotlin Fundamentals)  
    ArtSpaceApp/           (Project: Create an Art Space app)
```

- Add the instructor as a collaborator using the following email address:
rmaciaszczyk@zut.edu.pl
- Submit the repository link as a response to this assignment in **Microsoft Teams**.